

Performance Measures of Disks

Access Time

- **Definition:** Time from when a read/write request is issued to when data transfer begins.
- **Components:**
 - **Seek Time:** Time to reposition the arm over the correct track.
 - * Average seek time $\approx \frac{1}{2}$ of worst case seek time.
 - * Would be $\frac{1}{3}$ if all tracks had the same number of sectors and ignoring acceleration/deceleration.
 - * Typical values: **4–10 ms**.
 - **Rotational Latency:** Time for desired sector to rotate under head.
 - * Typical: **4–11 ms** (5400–15000 RPM disks).
 - * Average latency = $\frac{1}{2}$ of max latency.
 - **Overall Latency:** **5–20 ms** depending on disk model.

Data Transfer Rate

- Rate at which data can be retrieved from or stored to disk.
- Typical: **25–200 MB/s** (slower on inner tracks).
- **Disk Block:** Logical unit for storage allocation and retrieval.
 - Size: **4–16 KB**.
 - Smaller blocks $\Rightarrow$ more disk transfers.
 - Larger blocks $\Rightarrow$ more internal fragmentation.

Access Patterns

- **Sequential Access:**
 - Successive requests for successive blocks.
 - Seek required only for first block.
- **Random Access:**
 - Successive requests for blocks anywhere on disk.
 - Each access requires a seek $\Rightarrow$ slower performance.

I/O Operations per Second (IOPS)

- Number of random block reads a disk can support per second.
- Typical: **50–200 IOPS** for magnetic disks.

Reliability Measures

- **Mean Time To Failure (MTTF):**
 - Average continuous run time before a disk fails.
 - Typical: **3–5 years**.
 - Theoretical MTTF (new disks): **500,000–1,200,000 hours**.
- Example:
 - $MTTF = 1,200,000$ hours.
 - For 1000 disks, on average one disk fails every 1200 hours (≈ 50 days).
 - Equivalent to **7.3 disks/year**.
- **Annualized Failure Rate (AFR):**

$$AFR = \frac{365 \times 24}{MTTF} \times 100\%$$

- Example: $MTTF = 1,200,000$ hours $\Rightarrow$ $AFR = \mathbf{0.73\%}$.

MTTF and AFR Calculations

Definitions:

- **MTTF (Mean Time To Failure):** Average time a disk operates before failure.
- **AFR (Annualized Failure Rate):** Percentage of devices expected to fail in one year.

Hours in a year:

$$365 \times 24 = 8760 \text{ hours/year}$$

AFR Formula:

$$AFR = \frac{\text{Hours in a year}}{\text{MTTF}} \times 100\%$$

Example: MTTF = 1,200,000 hours

$$AFR = \frac{8760}{1,200,000} \times 100\% = 0.73\%$$

Failures in a system of 1000 disks:

$$\text{Average failure interval per disk} = \frac{MTTF}{N} = \frac{1,200,000}{1000} = 1200 \text{ hours} \approx 50 \text{ days}$$

$$\text{Failures per year} = \frac{8760 \times 1000}{1,200,000} \approx 7.3 \text{ disks/year}$$

Summary:

- One disk fails every 1200 hours on average.
- Approximately 7 disks fail per year in a 1000-disk system.
- AFR = 0.73%.

Summary Table of Disk Performance Measures

Measure	Typical Value / Notes
Seek Time	4–10 ms (avg $\approx 1/2$ worst case)
Rotational Latency	4–11 ms (5400–15000 RPM), avg = $1/2$
Overall Latency	5–20 ms depending on disk model
Data Transfer Rate	25–200 MB/s
Disk Block Size	4–16 KB
Access Patterns	Sequential / Random
IOPS (Random Access)	50–200 ops/sec
MTTF	500,000–1,200,000 hours (new disks)
AFR	$\approx 0.73\%$ (MTTF = 1,200,000 h)
Example Failure Rate	1 disk fails every 1200 h (1000 disks)

Table 1: Quick Reference: Disk Performance Measures